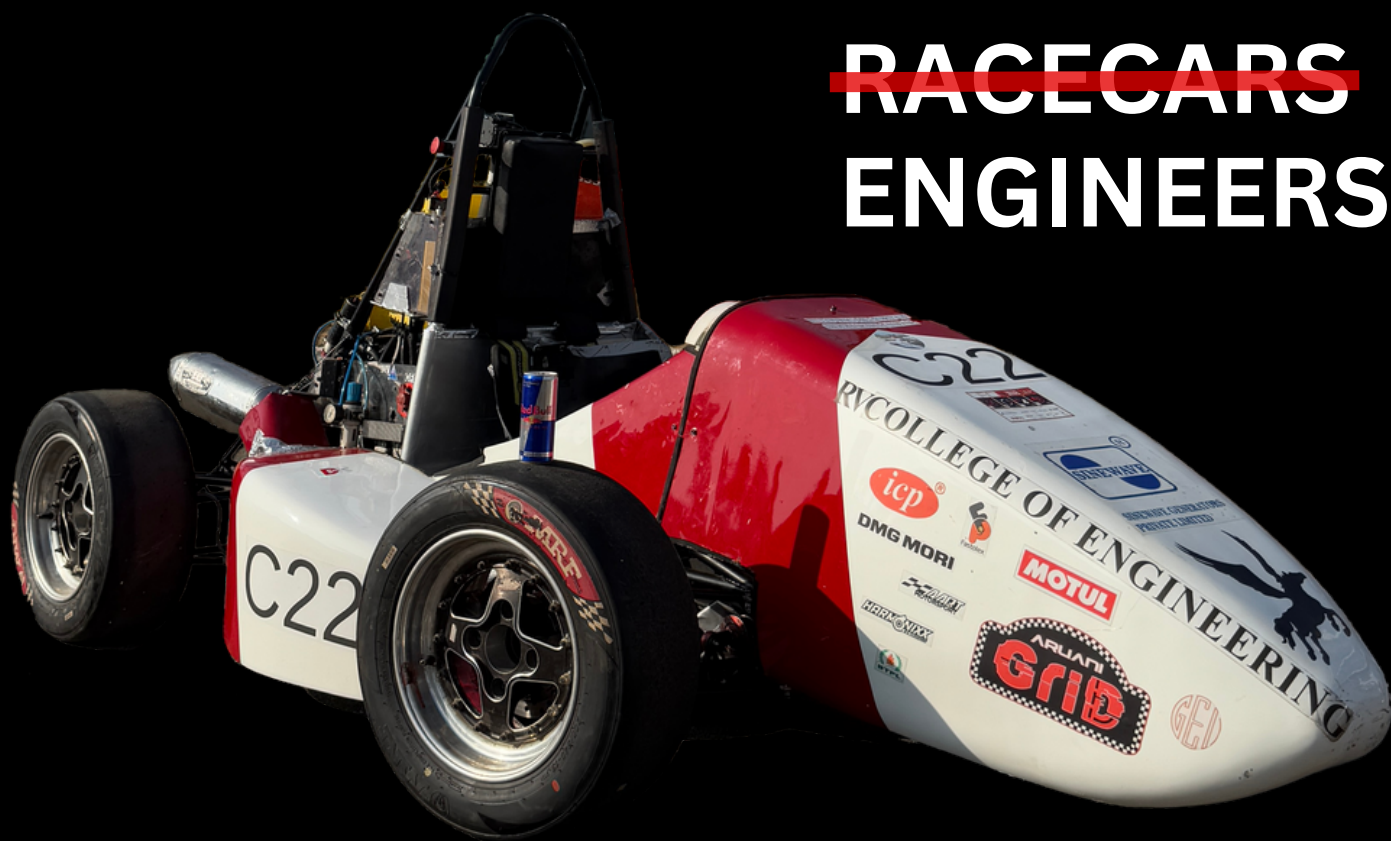




ashwa MOBILITY FOUNDATION

LIVE IT. LOVE IT **RACE IT.**

WE BUILD
~~RACECARS~~
ENGINEERS.



**FEBRUARY
2025
EDITION**

FORMULA BHARAT- THE IMPACT IT HAS ON YOUR LIFE



CONTENTS

SL NO.	TITLE	PAGE NO.
1	VISION AND MISSION	3
2	ALUMNI CORNER	5
3	TEAM LEAD SPEAKS	7
4	OUR JOURNEY IN FB'25	9
5	FUTURE OF NEWSLETTER	15
6	OUR SPONSORS	17
7	TEAM MEMBERS LIST	20



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**VISION
AND
MISSION**

GUIDING PURPOSE,
INSPIRING FUTURE.

MISSION AND VISION

VISION

- To be the national leader in all categories of Formula Student, to push the limits of engineering and compete in events at the highest level.
- To foster student mobility solutions in the future by developing sustainable prototypes across various domains.

MISSION

- Prepare students for the industry by designing and building world-class vehicles while facilitating industry interaction.
- Showcase India's engineering expertise in mobility on a global stage and provide talent opportunities for the industry.
- Develop cost-effective, sustainable transportation solutions for the masses.



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ALUMNI CORNER

CONNECTING MEMORIES,
BUILDING FUTURES!



ALUMNI CORNER

Ashwa Mobility Foundation has built a strong alumni network, with former members now excelling in leading multinational companies and entrepreneurial ventures. Since its establishment in 2003, Ashwa Mobility has provided students with hands-on experience in designing and manufacturing formula-style race cars, shaping them into skilled professionals ready for industry challenges.

Our alumni have made remarkable strides in the mobility and automotive sectors, applying the technical expertise and leadership skills they developed during their time with Ashwa. One such example is *Abhishek Chandrashekar, a 2015 graduate, who played a key role in the Chassis and Workshop Management subsystem. He went on to co-found Royal Brothers, a successful bike rental startup that has transformed urban mobility in South India.* His journey reflects the entrepreneurial mindset and problem-solving abilities that Ashwa Mobility instills in its members.

Beyond individual successes, our alumni remain an integral part of the Ashwa Mobility ecosystem, providing mentorship, career guidance, and industry insights to current members.

The achievements of our alumni stand as a testament to Ashwa Mobility's commitment to innovation and excellence. As we celebrate their contributions, we look forward to write to you about their inspiring stories on how Ashwa shaped their careers, straight from themselves. **Stay tuned as we share the journey our Alumni's from different walks of life.**



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**TEAM
LEAD
SPEAKS**

LEADING THE WAY: INSIGHTS FROM THE
FRONTLINES OF TEAM LEADERSHIP



TEAM CAPTAIN

RAVIKIRAN K

For me, leading Ashwa Racing has been one of the most challenging yet rewarding experiences of my life. *Every day, I discovered the ability to push myself harder than before, uncovering strengths I never knew I had.* I took on this role because I saw it as an opportunity to develop essential skills early on. However, being a Team Captain is far more demanding than I initially imagined.



Throughout this journey, I learned invaluable lessons about leadership, team dynamics, and resilience. The challenges I faced, the mistakes I made, and the small victories along the way have shaped my personality for the better. *Being a Team Captain means being the face of the team when no one else steps forward, making difficult decisions when no one else can, and persisting even when everything seems to be working against you.*

During my tenure, I encountered numerous difficulties, especially at the beginning, and often questioned my capabilities and place in the team. But with each passing day, I became more resilient, learning to push forward despite setbacks. I understood the importance of patience, mindfulness, and setting clear goals for the team—things that are much easier said than done. This role has undoubtedly helped me grow, both as a leader and as an individual.

Formula Bharat 2025 was an eye-opening experience and a true test of my ability to lead a team of talented individuals. While the results weren't what we had hoped for, I am grateful for the journey. *The challenges we faced have only fueled my hunger and persistence, strengthening my resolve to one day achieve the success I aspire to.*



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**THE JOURNEY
OF FORMULA
BHARAT 2025**

OVERCOMING CHALLENGES AND
SHOWCASING EXCELLENCE



DAY 0: GEARING UP FOR THE CHALLENGE

Every year, Formula Bharat brings together some of the brightest engineers in India to test their skills in designing, building, and racing formula-style vehicles. This year, it took place from January 22nd to 27th at the Kari Motor Speedway, Coimbatore and the team's participation reflected a strong commitment to technical excellence, teamwork, and problem-solving in a high-pressure environment.

Excitement filled the air as the team arrived at the venue and set up the pit. The team began with unloading the car and preparing the workspace. By 11:00 AM, registration was completed, and they dove straight into the crucial task of fine-tuning the prototype - XX5-C. The main focus was to resolve key mechanical aspects, including suspension adjustments and drivetrain assembly. However, during these checks, they identified a significant gap at the front-left rocker clamp, an issue that needed immediate attention. As time passed, preparations continued into the evening.



The team arranging the pits



Formula bharat 2025 Lineup

“Day 0 of Formula Bharat was going well as we set up the car, but towards the end of the day, we discovered a crack in the weld of the rocker clamp—possibly caused during transportation. This could have ended our competition, but we quickly assessed the situation and had it welded back the next day at the welding station in the hot pits area.”- Yashas B, Suspension and Admin.



Panoramic Photo Session



DAY 1: TECHNICAL HURDLES AND BREAK THROUGHS

The day began with a panoramic photoshoot before they rushed to a welder to address a hole in the chassis. After clearing the pre-technical inspection, new challenges emerged. Car's ground clearance issues, the bodywork and side-pods scraping the track, demanded a swift redesign of the pull rods. While addressing a rear brake line leak, the bleed screw sheared off, forcing to replace the calliper. Despite these setbacks, the team remained determined. In the mechanical technical inspection, the inspectors flagged several concerns, like the front-right wheel wobble, Positive stops for clutch and accelerator pedals, overlap check for the aluminum firewall, and requirement for aluminum tape on the inside of bodyworks.



Team Fine tuning the car



DAY 2: PERSISTENCE PAYS OFF

Determined to get our car race-ready, the technical issues that were identified the previous day were tackled. Meanwhile, the Cost and Manufacturing Report was presented at the Event Finals, marking a crucial milestone in the competition. After meticulous adjustments and rigorous re-evaluations, the mechanical technical inspection was cleared on the second attempt, a testament to the team's unwavering dedication and problem-solving abilities



Passing Mechanical Inspection



At the Cost and Manufacturing Finals



DAY 3: TRIUMPHS AND SETBACKS

The day started with the tilt test, an essential evaluation to ensure vehicle stability. Initially, leaks in brake fluid and engine oil were detected, but after swift repairs, the test was cleared. The noise test was also cleared bringing the team one step closer to the dynamic events. However, during the brakes test, right half shaft failed at the wheel end during the third launch. It was quickly replaced with a redesigned version, only to face another setback—the left half shaft failed at the CV bell end cap. Recognizing a pattern, a complete redesign and manufacturing process was initiated overnight.



RZ-XX5-C at the Tilt test



DAY 4: THE RACE AGAINST TIME



Clearing brakes test after multiple attempts

With no time to spare, the team manufactured four new half shafts using an alternative material and heat treatment processes. Two of these shafts were installed, and the brake test was reattempted. The right shaft failed again at the wheel end. Another replacement was made, but yet again, the right shaft succumbed to failure. In a last-ditch effort, they replaced the shafts with an older design, finally clearing the brakes test. Even though it was too late for the dynamic events, spirits were lifted as they secured 2nd place in the Cost Event Finals—a proud achievement for the team.

"To see the shafts break repeatedly, was worrying to say the least. When the shaft broke, we wasted no time and put all our efforts in replacing the shaft, and getting the car back on the track, only for it to return back with another broken shaft. Handling the failure was tough for all of us, to see the efforts you put, not yield the results you want, truly saddening but you learn more through failure, through mistakes and through consequences, than through success. Whether about the car, about engineering, or simply about ourselves, those shafts taught us a lot." -Adithya Ranjith, Drivetrain and HR



Team winning 2nd place in Cost Event Finals



DAY 5: ENDURANCE AND THE FINAL BATTLE

With renewed energy, we entered the endurance event. The car performed well initially, but disaster struck on the 7th lap—the left shaft failed, forcing withdrawal from the race. While this was not the outcome that we hoped, the resilience and adaptability throughout the event stood as a testament to their dedication and skill.



The entire team at FB'25



At the Endurance event

“Despite hurdles, we successfully built the XX5-C prototype and competed in Formula Bharat 2025. The competition was an incredible experience, and Ashwa taught me problem-solving, resilience, and pushing my limits. I will always remember the countless days and nights spent in the workshop building the car. For me the journey is its own reward, I could have never asked for a better team than this one.” – Chinmay, Chief Engineer.



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**WHAT'S
COMING
NEXT**

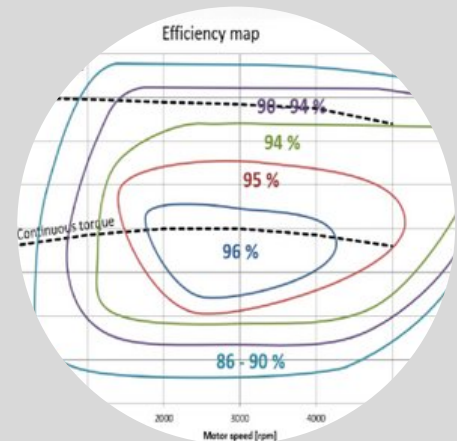
BRIDGING THEORY AND
REAL-WORLD ENGINEERING

As we continue pushing the boundaries of innovation, the upcoming editions of the Ashwa Mobility Foundation newsletter will feature exciting insights and in-depth discussions. Here's a glimpse of what's on the way!

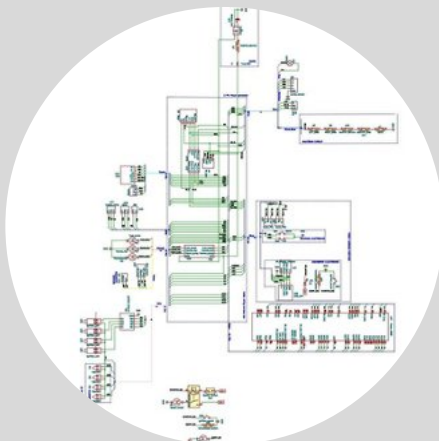


KNOWLEDGE TRANSFER: UNLOCKING MECHANICAL CONCEPTS

Have you ever wondered about the intricate mechanics behind our race cars? In the upcoming edition, we'll explore a fundamental mechanical concept that plays a crucial role in vehicle design. Whether you're an aspiring engineer or simply curious about the technology that drives innovation, this section will break it down in an engaging and easy-to-understand way!



KNOWLEDGE TRANSFER: UNLOCKING ELECTRICAL CONCEPTS



Beyond mechanics, our vehicles rely on complex electrical systems that enhance performance and efficiency. In the next issue, we will dive into an essential electrical engineering principle used in our design. Expect simplified explanations, real-world applications, and insights into how our electrical team optimizes the car's power system!

This is just a glimpse of the exciting content coming your way. Stay tuned as we continue to explore, innovate, and share the journey of Ashwa Mobility!



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AKSHAYA MOTORS





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**TEAM
DIRECTORY
(2024-2025)**

DRIVEN BY PASSION, UNITED BY ENGINEERING—
THIS IS ASHWA RACING!

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Chinmay G Vanahalli	Chief Engineer
Anirudh KN	Chassis and Workshop Management
Sumeeth Anil Jadhav	Chassis and Workshop Management
Ipparla Ashakiran	Brakes and Logistics
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